# 1 INTRODUCTION TO SEMICONDUCTORS

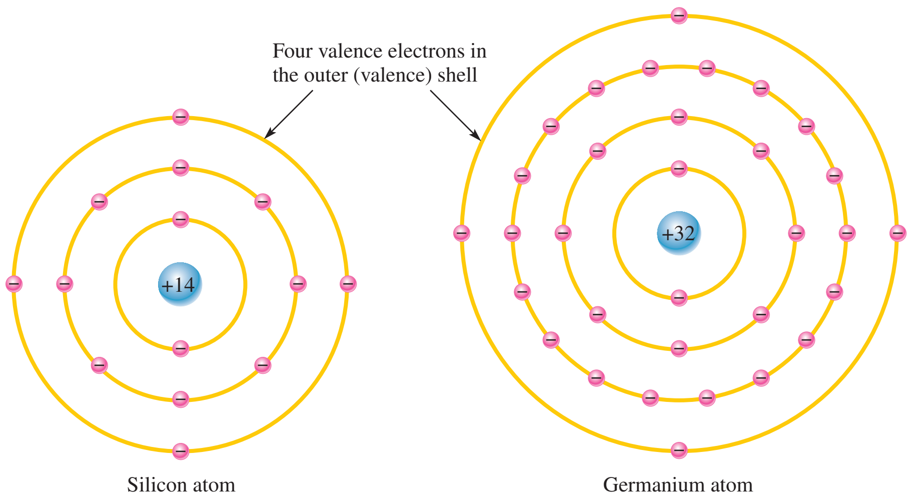
In this section, atomic theory is discussed in relation to semiconductors that are used in devices such as diodes and transistors.

After completing this section, you should be able to

- **Understand the basic structure of semiconductors and how they conduct current**
  - Describe the atomic structure of silicon and germanium
  - Discuss covalent bonding in silicon
  - Explain how current occurs with electrons and holes in a semiconductor
  - Describe the properties of *n*-type and *p*-type semiconductors

## Silicon and Germanium Atoms

Two types of semiconductive materials are **silicon** and germanium. Both the silicon and the germanium atoms have four valence electrons. These atoms differ in that silicon has 14 protons in its nucleus and germanium has 32. Figure 1 shows a representation of the atomic structure for both materials.




▶ FIGURE 1

Diagrams of the silicon and germanium atoms.

The valence electrons in germanium are in the fourth shell while the ones in silicon are in the third shell, closer to the nucleus. This means that the germanium valence electrons are at higher energy levels than those in silicon and, therefore, require a smaller amount of additional energy to escape from the atom. This property makes germanium more unstable than silicon at high temperatures, which is the main reason silicon by far is the most widely used semiconductive material.

Although silicon is used in virtually all integrated circuits, there are some limited applications for germanium. Recent advances have created a "porous germanium" that has a nanoscale honeycombed lattice arrangement of germanium atoms with a huge surface area for a given amount of material. This has opened some interesting possibilities for applications of germanium in sensitive detectors and solar cells. Because germanium has

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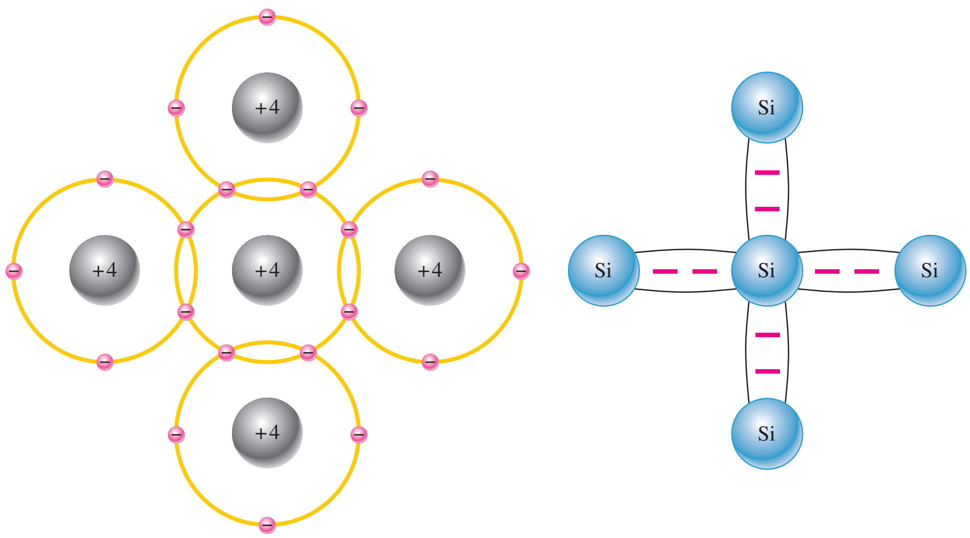
only limited applications, we will assume silicon is the semiconductor used in discussions of electronic devices in this text.

## Atomic Bonding

When certain atoms combine to form a solid material, they arrange themselves in a fixed pattern called a **crystal**. The atoms within the silicon crystal structure are held together by **covalent** bonds, which are created by the sharing of the valence electrons of each atom.

Figure 2 shows how each silicon (Si) atom positions itself with four adjacent atoms to form a silicon crystal. A silicon atom with its four valence electrons shares an electron with each of its four neighbors. This effectively creates eight valence electrons for each atom and produces a state of chemical stability. Also, this sharing of valence electrons produces the covalent bonds that hold the atoms together; each shared electron is attracted equally by two adjacent atoms. An **intrinsic** crystal is one that has no impurities.




(a) The center silicon atom shares an electron with each of the four surrounding silicon atoms, creating a covalent bond with each. The surrounding atoms are in turn bonded to other atoms, and so on.

(b) Bonding diagram. The red negative signs represent the shared valence electrons.

FIGURE 2

Covalent bonds in a silicon crystal. The actual crystal is three-dimensional.

## Conduction Electrons and Holes

An energy band diagram is shown in Figure 3 for a silicon crystal with only unexcited atoms (no external energy such as heat). This condition occurs *only* at a temperature of absolute 0 Kelvin.

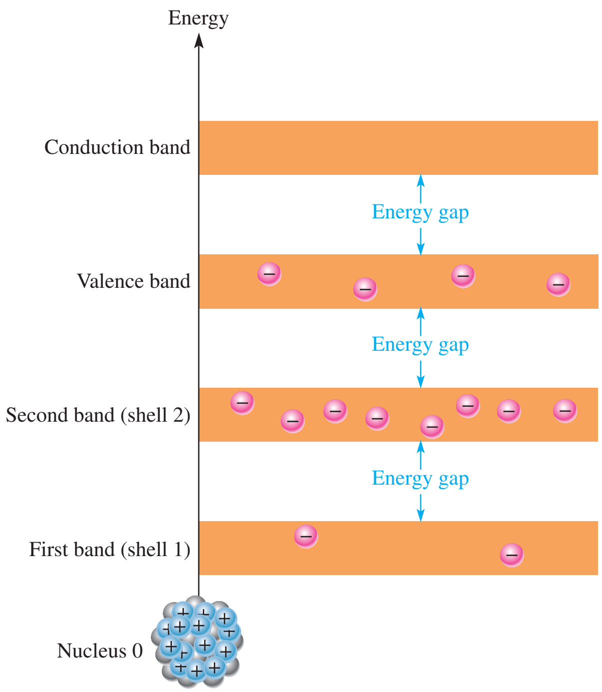
An intrinsic (pure) silicon crystal at room temperature has sufficient heat (thermal) energy for some valence electrons to jump the gap from the valence band into the conduction band, becoming free electrons. This situation is illustrated in the energy diagram of Figure 4(a) and in the bonding diagram of Figure 4(b).

When an electron jumps to the conduction band, a vacancy is left in the valence band within the crystal. This vacancy is called a **hole**. For every electron raised to the conduction band by external energy, there is one hole left in the valence band, creating what is called an *electron-hole pair*. **Recombination** occurs when a conduction-band electron loses energy and falls back into a hole in the valence band.

To summarize, a piece of intrinsic silicon at room temperature has, at any instant, a number of conduction-band (free) electrons that are unattached to any atom and are essentially drifting randomly throughout the material. Also, an equal number of holes are created in the valence band when these electrons jump into the conduction band. This is illustrated in Figure 5.

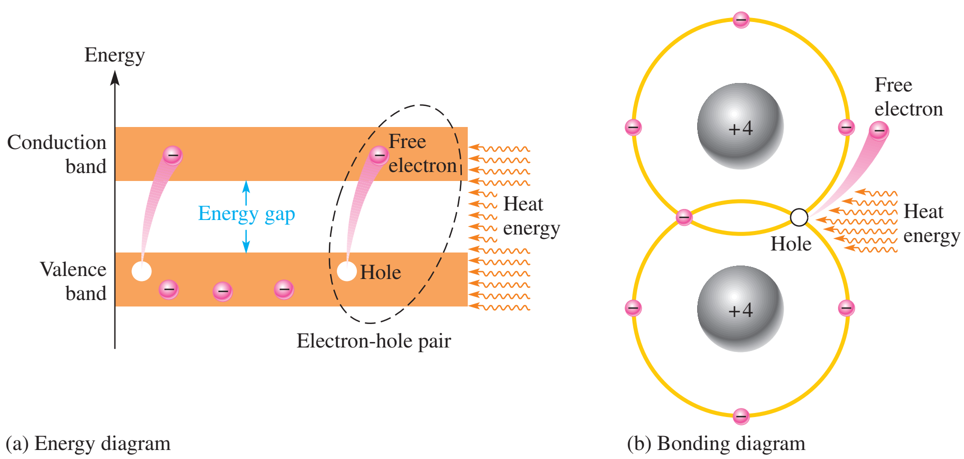
▶ **FIGURE 3**

Energy band diagram for a pure silicon crystal with unexcited atoms. There are no electrons in the conduction band.



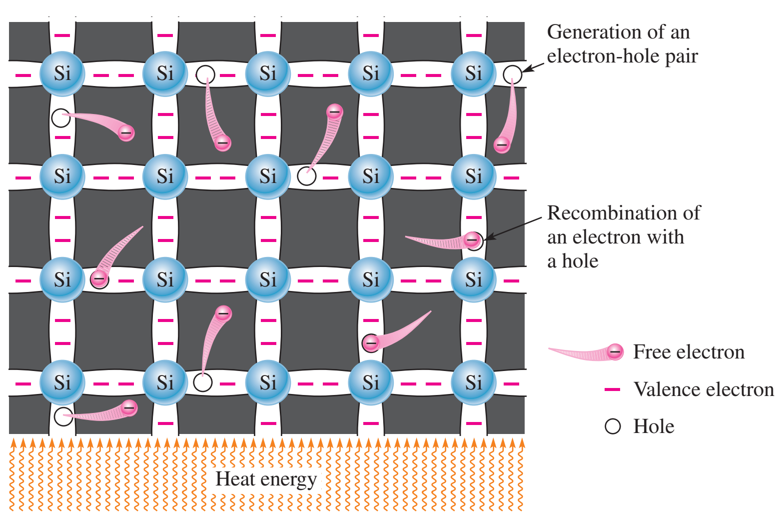

▶ **FIGURE 4**

Creation of electron-hole pairs in a silicon crystal. Electrons in the conduction band are free electrons.




▶ **FIGURE 5**

Electron-hole pairs in a silicon crystal. Free electrons are being generated continuously while some recombine with holes.



## Electron and Hole Current

When a voltage is applied across a piece of silicon, as shown in Figure 6, the thermally generated free electrons in the conduction band, which are free to move randomly in the crystal structure, are now easily attracted toward the positive end. This movement of free electrons is one type of current in a semiconductive material and is called *electron current*.

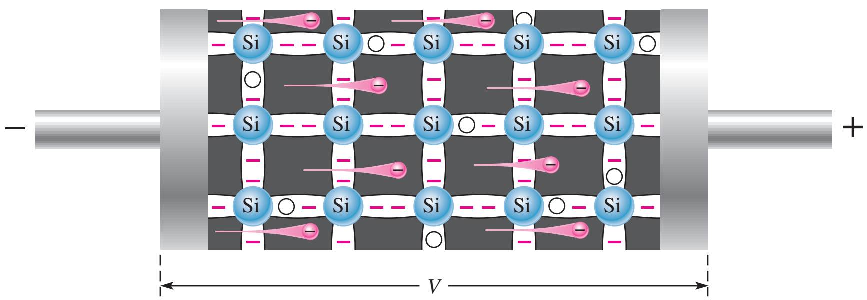
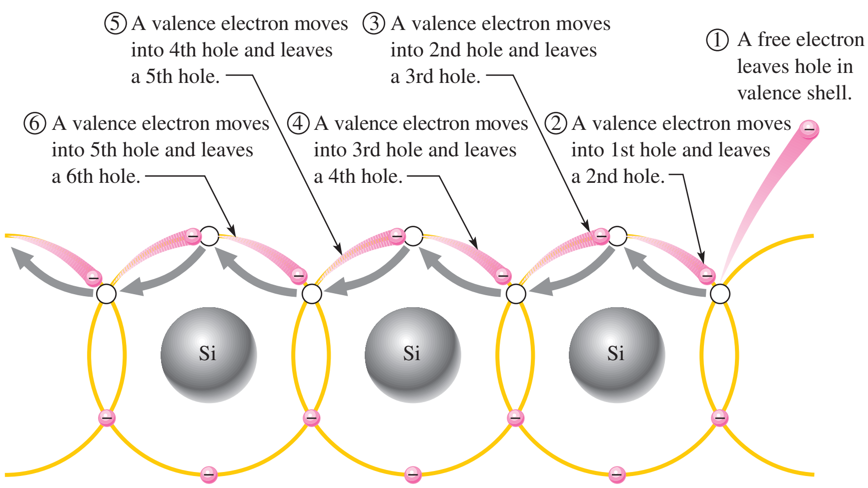



**FIGURE 6**

Free electron current in intrinsic silicon is produced by the movement of thermally generated free electrons in the conduction band.

Another type of current occurs at the valence level, where the holes created by the free electrons exist. Electrons remaining in the valence band are still attached to their atoms and are not free to move randomly in the crystal structure. However, a valence electron can move into a nearby hole, with little change in its energy level, thus leaving another hole where it came from. The hole has effectively, although not physically, moved from one place to another in the crystal structure, as illustrated in Figure 7. This current is called *hole current*.




When a valence electron moves left to right to fill a hole while leaving another hole behind, the hole has effectively moved from right to left. Gray arrows indicate effective movement of a hole.

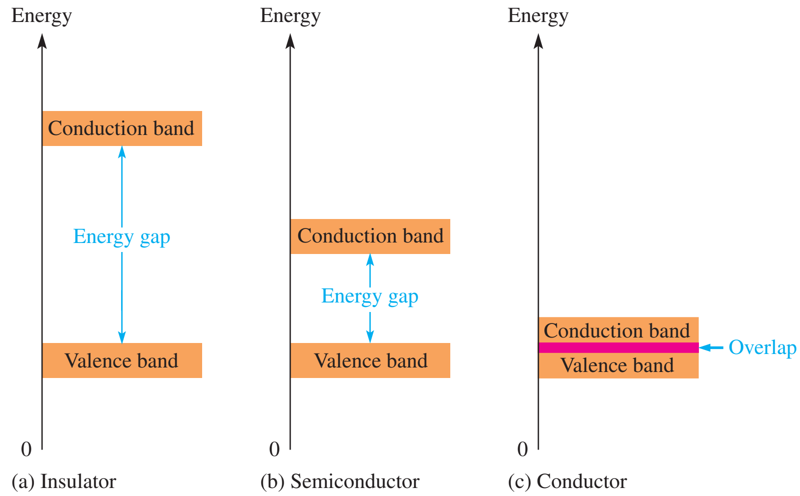
**FIGURE 7**

Hole current in intrinsic silicon.

## Comparison of Semiconductors to Conductors and Insulators

In an **intrinsic semiconductor**, there are relatively few free electrons, so semiconductors are not very useful in their intrinsic states. Pure semiconductive materials are neither insulators nor good conductors because current in a material depends directly on the number of free electrons.

A comparison of the energy bands in Figure 8 for insulators, semiconductors and conductors shows the essential differences among them regarding conduction. The energy gap for an insulator is so wide that hardly any electrons acquire enough energy to jump into the conduction band. The valence band and the conduction band in a conductor (such as copper) overlap so that there are always many conduction electrons, even without the application of external energy. A semiconductor, as Figure 8(b) shows, has an energy gap that is much narrower than that in an insulator.




▲ FIGURE 8

Energy diagrams for the three types of materials.

## *N*-Type and *P*-Type Semiconductors

Semiconductive materials do not conduct current well and are of little value in their intrinsic state. This is because of the limited number of free electrons in the conduction band and holes in the valence band. Intrinsic silicon must be modified by increasing the free electrons and holes to increase its conductivity and make it useful in electronic devices. This is done by adding impurities to the intrinsic material as you will learn in this section. Two types of extrinsic (impure) semiconductive materials, *n*-type and *p*-type, are the key building blocks for all types of electronic devices.

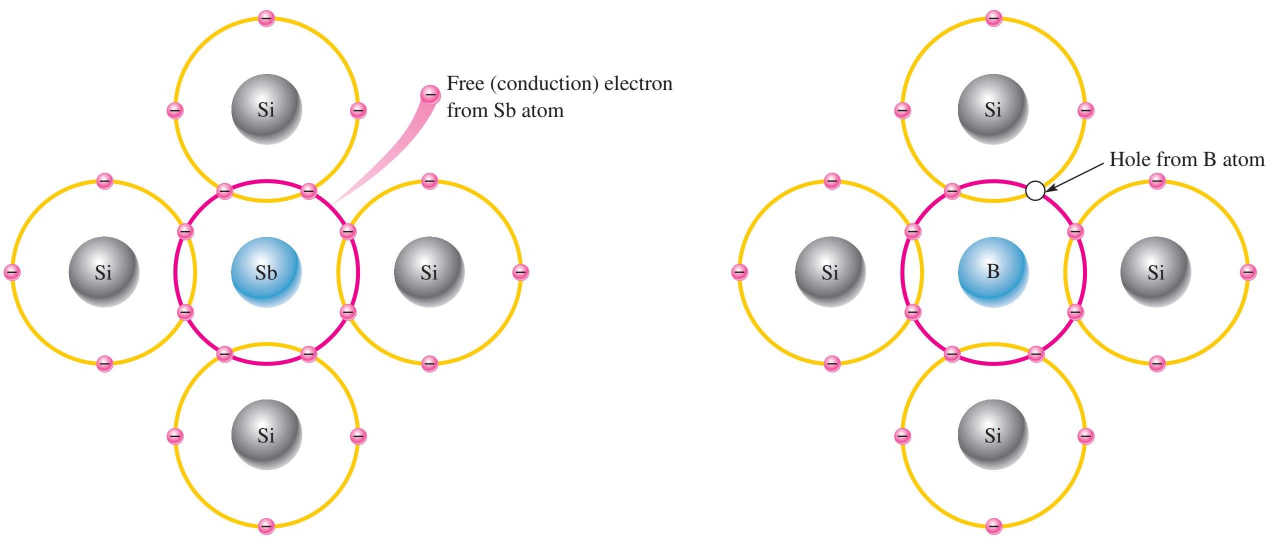
***Doping*** The conductivities of silicon can be drastically increased and controlled by the addition of impurities to the intrinsic (pure) semiconductive material. This process, called **doping**, increases the number of current carriers (electrons or holes). The two categories of impurities are *n-type* and *p-type.*

***N-Type Semiconductor*** To increase the number of conduction-band electrons in intrinsic silicon, **pentavalent** impurity atoms are added. These are atoms with five valence electrons, such as arsenic (As), phosphorus (P), and antimony (Sb) and are known as donor atoms because they provide an extra electron to the semiconductor's crystal structure.

As illustrated in Figure 9(a), each pentavalent atom (antimony, in this case) forms covalent bonds with four adjacent silicon atoms. Four of the antimony atom's valence electrons are used to form the covalent bonds with silicon atoms, leaving one extra electron. This extra electron becomes a conduction electron because it is not attached to any atom. The number of conduction electrons can be controlled by the number of impurity atoms added to the silicon.

Since most of the current carriers are electrons, silicon doped in this way is an *n*-type semiconductor (the *n* stands for the negative charge on an electron). The electrons are called the **majority carriers** in *n*-type material. Although the majority of current carriers in *n*-type material are electrons, there are some holes. These holes are *not* produced by the addition of the pentavalent impurity atoms. Holes in an *n*-type material are called **minority carriers**.

***P-Type Semiconductor*** To increase the number of holes in intrinsic silicon, **trivalent** impurity atoms are added. These are atoms with three valence electrons, such as aluminum (Al), boron (B), and gallium (Ga) and are known as acceptor atoms because they leave a hole in the semiconductor's crystal structure.




(a) Pentavalent impurity atom in a silicon crystal. An antimony (Sb) impurity atom is shown in the center. The extra electron from the Sb atom becomes a free electron.

(b) Trivalent impurity atom in a silicon crystal. A boron (B) impurity atom is shown in the center.

▲ **FIGURE 9**

Impurity atoms.

As illustrated in Figure 9(b), each trivalent atom (boron, in this case) forms covalent bonds with four adjacent silicon atoms. All three of the boron atom's valence electrons are used in the covalent bonds; and, since four electrons are required, a hole is formed with each trivalent atom. The number of holes can be controlled by the amount of trivalent impurity added to the silicon.

Since most of the current carriers are holes, silicon doped with trivalent atoms is a *p*-type semiconductor. Holes can be thought of as positive charges. The holes are the majority carriers in *p*-type material. Although the majority of current carriers in *p*-type material are holes, there are also a few free electrons that are created when electron-hole pairs are thermally generated. These few free electrons are *not* produced by the addition of trivalent impurity atoms. Electrons in *p*-type material are the minority carriers.

**SECTION 1 CHECKUP**
Answers are at the end of the chapter.

1. How are covalent bonds formed?
2. What is meant by the term *intrinsic*?
3. What is a crystal?
4. Effectively, how many valence electrons are there in each atom within a silicon crystal?
5. In the atomic structure of a semiconductor, within which energy band do free electrons exist? Within which energy band do valence electrons exist?
6. How are holes created in an intrinsic semiconductor?
7. Why is current established more easily in a semiconductor than in an insulator?
8. How is an *n*-type semiconductor formed?
9. How is a *p*-type semiconductor formed?
10. What are majority carriers?

## 2 THE DIODE

If you take a block of silicon and dope half of it with a trivalent impurity and the other half with a pentavalent impurity, a boundary called the ***pn* junction** is formed between the resulting *p*-type and *n*-type portions of a semiconduction diode.

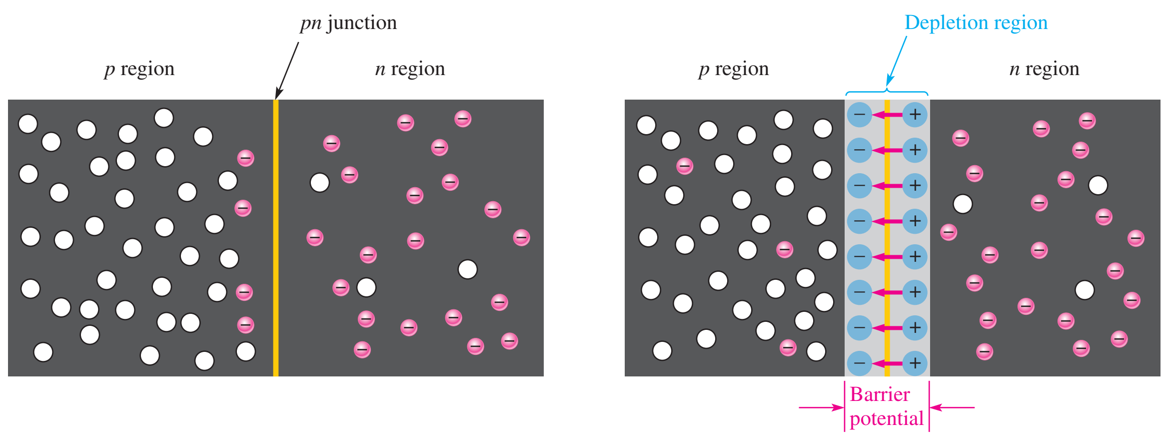
After completing this section, you should be able to

- **Describe the characteristics and biasing of a diode**
  - Define *pn* junction
  - Discuss the depletion region in a diode
  - Define *barrier potential*
  - Discuss forward bias
  - Discuss reverse bias
  - Define *reverse breakdown*

There is no movement of electrons (current) through a diode at equilibrium. The primary usefulness of the diode is its ability to allow current in only one direction and to prevent current in the other direction as determined by the bias. In electronics, **bias** refers to the use of a dc voltage to establish certain operating conditions for an electronic device. There are two practical bias conditions for a diode: forward and reverse. Either of these conditions is created by application of a sufficient external voltage of the proper polarity across the *pn* junction.

### Formation of the Depletion Region in a Diode

A **diode** consists of an *n* region and a *p* region separated by a *pn* junction, as illustrated in Figure 10. The *n* region has many conduction electrons, and the *p* region has many holes. With no external voltage, the conduction electrons in the *n* region are randomly drifting in all




(a) At the instant of junction formation, free electrons in the *n* region near the *pn* junction begin to diffuse across the junction and fall into holes near the junction in the *p* region.

(b) For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the *n* region and a negative charge is created in the *p* region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion.

▲ FIGURE 10

Formation of the depletion region in a diode.

directions. At the instant of junction formation, some of the electrons near the junction drift across into the *p* region and recombine with holes near the junction, as shown in part (a).

For each electron that crosses the junction and recombines with a hole, a pentavalent atom is left with a net positive charge in the *n* region near the junction, making it a positive ion. Also, when the electron recombines with a hole in the *p* region, a trivalent atom acquires net negative charge, making it a negative ion.

As a result of this recombination process, a large number of positive and negative ions builds up near the *pn* junction. As this buildup occurs, the electrons in the *n* region must overcome both the attraction of the positive ions and the repulsion of the negative ions in order to migrate into the *p* region. Thus, as the ion layers build up, the area on both sides of the junction becomes essentially depleted of any conduction electrons or holes and is known as the *depletion region.* This condition is illustrated in Figure 10(b). When an equilibrium condition is reached, the depletion region has widened to a point where no more electrons can cross the *pn* junction.

The existence of the positive and negative ions on opposite sides of the *pn* junction creates a barrier potential across the depletion region, as indicated in Figure 10(b). The **barrier potential**, $V_B$, is the amount of voltage required to move electrons through the depletion region. At 25°C, it is approximately 0.7 V for silicon and 0.3 V for germanium. As the junction temperature increases, the barrier potential decreases, and vice versa.

## Biasing a Diode

***Forward Bias*** **Forward bias** is the condition that permits current through a diode. Figure 11 shows a dc voltage connected in a direction to forward-bias the *pn* junction. Notice that the negative terminal of the $V_{BIAS}$ source is connected to the *n* region, and the positive terminal is connected to the *p* region.

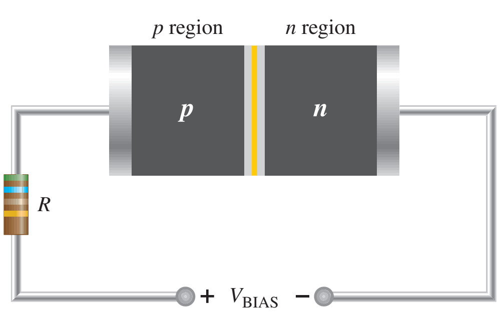


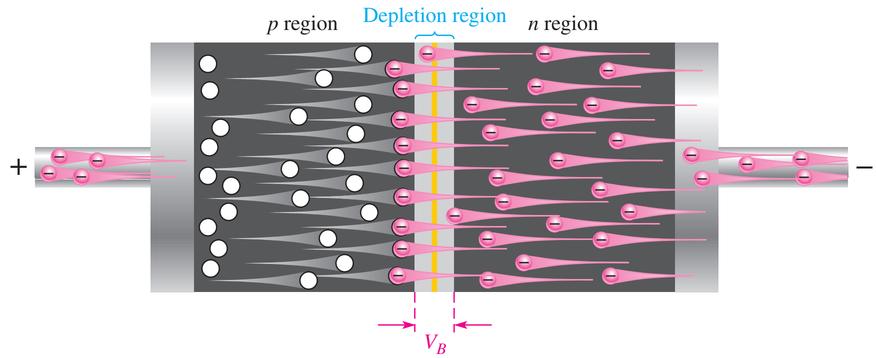

**FIGURE 11**

Forward-bias connection. The resistor limits the forward current in order to prevent damage to the diode.

A discussion of the basic operation of forward bias follows: The negative terminal of the bias-voltage source pushes the conduction-band electrons in the *n* region toward the *pn* junction, while the positive terminal pushes the holes in the *p* region also toward the *pn* junction. Recall that like charges repel each other.

When it overcomes the barrier potential ($V_B$), the external voltage source provides the *n*-region electrons with enough energy to penetrate the depletion region and move through the junction, where they combine with the *p*-region holes. As electrons leave the *n*-region, more flow in from the negative terminal of the bias-voltage source. Thus, current through the *n* region is formed by the movement of conduction electrons (majority carriers) toward the *pn* junction.

Once the conduction electrons enter the *p* region and combine with holes, they become valence electrons. Then they move as valence electrons from hole to hole toward the positive connection of the bias-voltage source. The movement of these valence electrons is the same as the movement of holes in the opposite direction. Thus, current in the *p* region is formed by the movement of holes (majority carriers) toward the *pn* junction. Figure 12 illustrates current in a forward-biased diode.



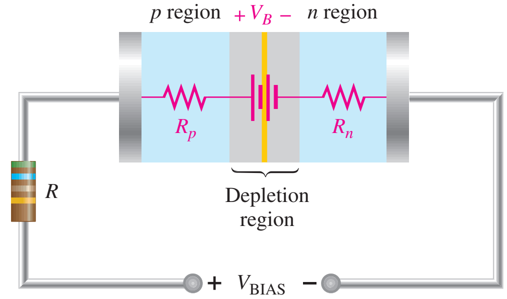

▲ **FIGURE 12**

**Current in a forward-biased diode.**

The effect of the barrier potential in the depletion region is to oppose forward bias. This is because the negative ions near the junction in the *p* region tend to prevent electrons from moving through the junction into the *p* region. You can think of the barrier potential effect as simulating a small battery connected in a direction to oppose the forward-bias voltage, as shown in Figure 13. The resistances $R_p$ and $R_n$ represent the dynamic resistances of the *p* and *n* materials.

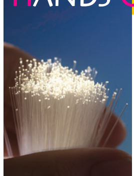
▶ **FIGURE 13**

**Barrier potential and dynamic resistance equivalent for a diode.**




The external bias voltage must overcome the effect of the barrier potential before the diode conducts, as illustrated in Figure 14. Conduction occurs at approximately 0.7 V for silicon. Once the diode is conducting in the forward direction, the voltage drop across it remains at approximately the barrier potential and changes very little with changes in forward current ($I_F$), as illustrated in Figure 14.

**HANDS ON TIP**

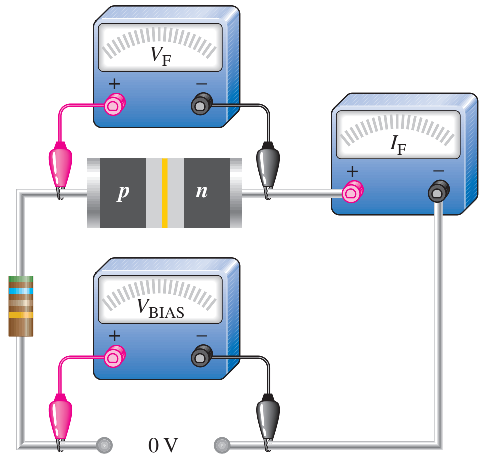


A special type of diode that makes use of the depletion-region capacitance is the *varactor*. It is used as a variable capacitor in which the capacitance is controlled by the reverse-bias voltage. By increasing or decreasing the reverse voltage, you can increase or decrease the capacitance.

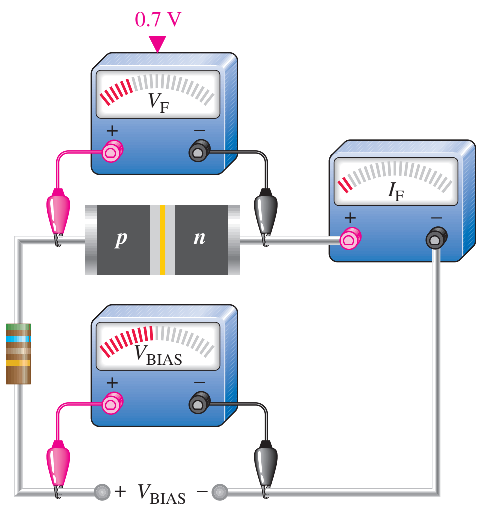
***Reverse Bias*** **Reverse bias** is the condition that prevents current through the diode. Figure 15(a) shows a dc voltage source connected to reverse-bias the diode. Notice that the negative terminal of the $V_{BIAS}$ source is connected to the *p* region and the positive terminal is connected to the *n* region.

A discussion of the basic operation for reverse bias follows: The negative terminal of the bias-voltage source attracts holes in the *p* region away from the *pn* junction, while the positive terminal also attracts electrons away from the *pn* junction. As electrons and holes move away from the *pn* junction, the depletion region widens; more positive ions are created in the *n* region, and more negative ions are created in the *p* region, as shown in Figure 15(b). The initial flow of majority carriers away from the *pn* junction is called *transient current* and lasts only for a very short time upon application of reverse bias.

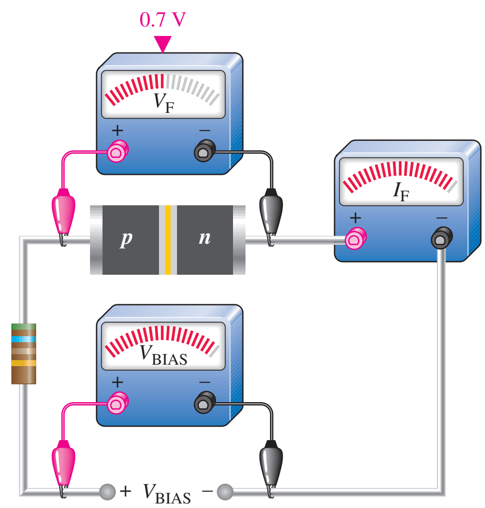
The depletion region widens until the potential difference across it equals the external bias voltage. At this point, the holes and electrons stop moving away from the *pn* junction, and majority current ceases, as indicated in Figure 15(c).




(a) No bias voltage. The *pn* junction of the diode is at equilibrium.




(b) Small forward-bias voltage ($V_F < 0.7$ V), very small forward current.




(c) Forward voltage reaches and remains at approximately 0.7 V. Forward current continues to increase as the bias voltage is increased.

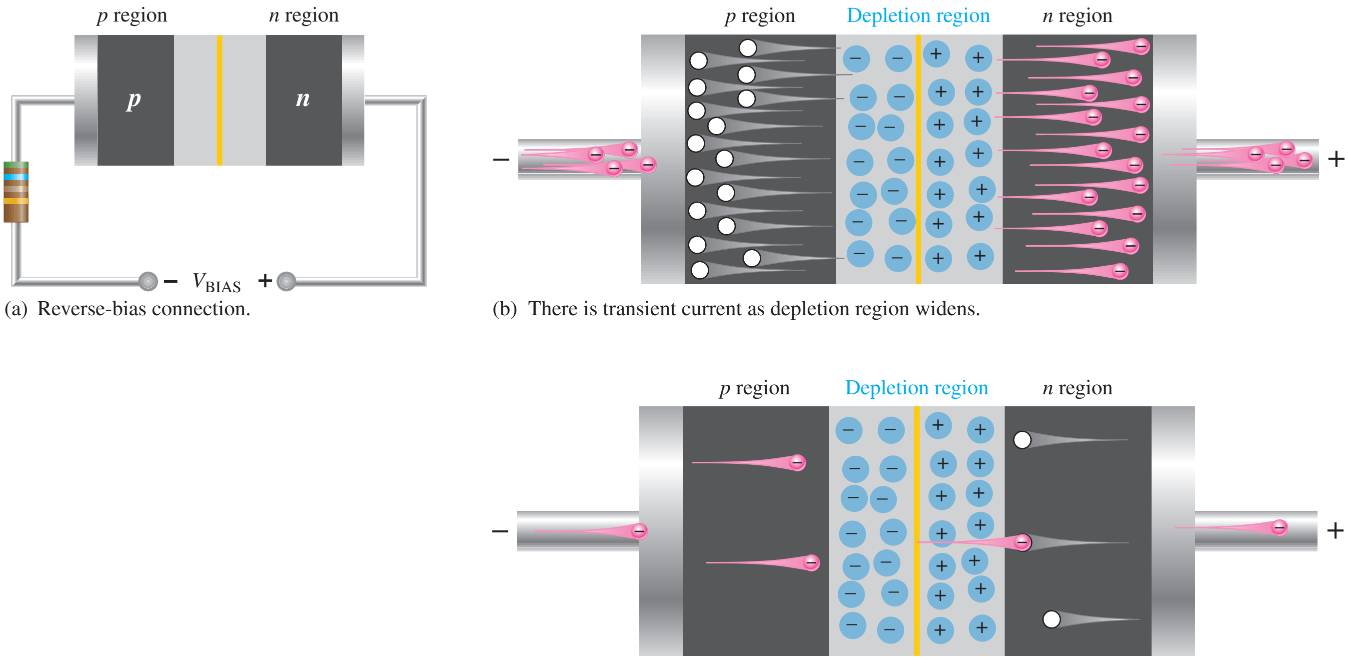
▲ **FIGURE 14**

**Illustration of diode operation under forward-bias conditions.**

When the diode is reverse-biased, the depletion region effectively acts as an insulator between the layers of oppositely charged ions, forming an effective capacitance. Since the depletion region widens with increased reverse-biased voltage, the capacitance decreases, and vice versa. This internal capacitance is called the *depletion-region capacitance.*

As you have learned, majority current very quickly becomes zero when reverse bias is applied. There is, however, a very small current produced by minority carriers during reverse bias. This current is typically in the $\mu$A or nA range. A relatively small number of thermally produced electron-hole pairs exist in the depletion region. Under the influence of the external voltage, some electrons manage to diffuse across the *pn* junction before recombination. This process establishes a small minority carrier current throughout the material.

The reverse current is dependent primarily on the junction temperature and not on the amount of reverse-biased voltage. A temperature increase causes an increase in reverse current.




(a) Reverse-bias connection.

(b) There is transient current as depletion region widens.

(c) Majority current ceases when barrier potential equals bias voltage. There is an extremely small reverse current due to minority carriers.

▲ **FIGURE 15**

Illustration of reverse bias.

***Reverse Breakdown*** If the external reverse-bias voltage is increased to a large enough value, **reverse breakdown** occurs. The following describes what happens: Assume that one minority conduction-band electron acquires enough energy from the external source to accelerate it toward the positive end of the diode. During its travel, it collides with an atom and imparts enough energy to knock a valence electron into the conduction band. There are now two conduction-band electrons. Each will collide with an atom, knocking two more valence electrons into the conduction band. There are now four conduction-band electrons which, in turn, knock four more into the conduction band. This rapid multiplication of conduction-band electrons, known as an *avalanche effect,* results in a rapid buildup of reverse current.

Most diodes normally are not operated in reverse breakdown and can be damaged if they are. However, a particular type of diode known as a zener diode (discussed in Section 6) is specially designed for reverse-breakdown operation.

**SECTION 2 CHECKUP**

1. What is a *pn* junction?
2. When *p* and *n* regions are joined, a depletion region forms. Describe the characteristics of the depletion region.
3. The barrier potential for silicon is greater than for germanium. (True or False)
4. What is the barrier potential for silicon at 25°C?
5. Name the two bias conditions.
6. Which bias condition produces majority carrier current?
7. Which bias condition produces a widening of the depletion region?
8. Minority carriers produce the current during reverse breakdown. (True or False)

## 3 DIODE CHARACTERISTICS

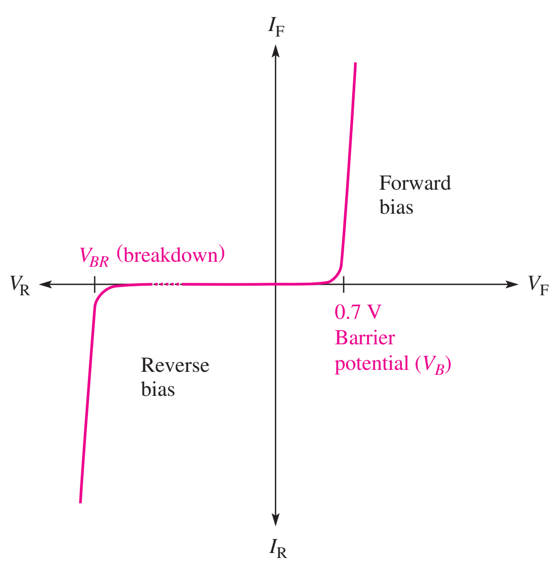
As you learned in the last section, a diode is a semiconductive device made with a single *pn* junction. A diode conducts current when it is forward-biased when the bias voltage exceeds the barrier potential. A diode prevents current when it is reverse-biased at less than the breakdown voltage.

After completing this section, you should be able to

- **Describe the basic diode characteristics**
  - Explain the diode *V-I* characteristic curve
  - Recognize the standard diode symbol
  - Test a diode with an ohmmeter
  - Describe three diode approximations

### Diode Characteristic Curve

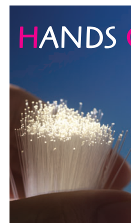
Figure 16 is a graph of diode voltage versus current, known as a *V-I* characteristic curve. The upper right quadrant of the graph represents the forward-biased condition. As you can see, there is very little forward current ($I_F$) for forward voltages ($V_F$) below the barrier potential. As the forward voltage approaches the value of the barrier potential, the current begins to increase. Once the forward voltage reaches the barrier potential, the current increases drastically and must be limited by a series resistor. *The voltage across the forward-biased diode remains approximately equal to the barrier potential.*



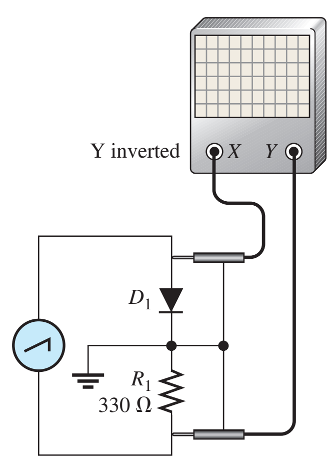

▲ FIGURE 16

General diode *V-I* characteristic curve.

HANDS ON TIP



The diode forward *V-I* characteristic curve can be plotted on the oscilloscope using the circuit shown below. Channel 1 senses the voltage across the diode and channel 2 senses a signal that is proportional to the current. The scope must be in the *X-Y* mode. The signal generator provides a 5 Vpp sawtooth or triangular waveform at 50 Hz, and its ground must not be the same as the scope ground. Channel 2 (the Y channel) must be inverted for the displayed curve to be properly oriented.

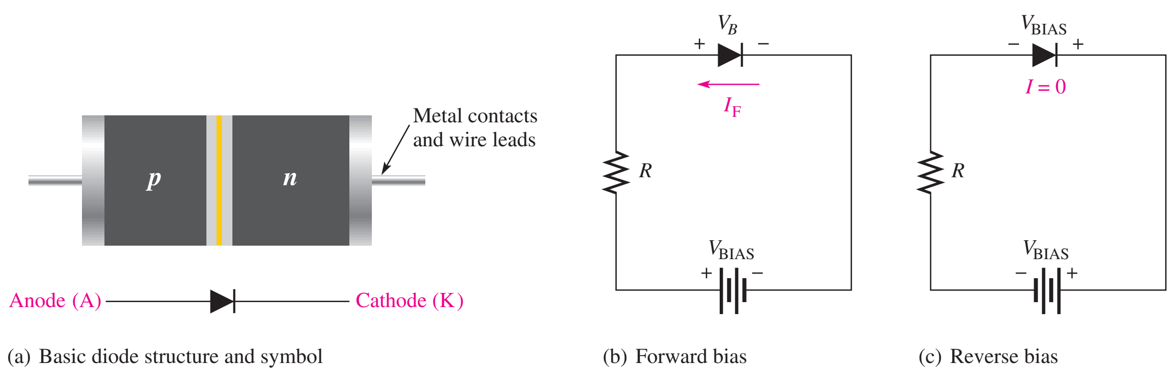



The lower left quadrant of the graph represents the reverse-biased condition. As the reverse voltage ($V_R$) increases to the left, the current remains near zero until the breakdown voltage ($V_{BR}$) is reached. When breakdown occurs, there is a large reverse current which, if not limited, can destroy the diode. Typically, the breakdown voltage is greater than 50 V for most rectifier diodes. Remember that most diodes should not be operated in reverse breakdown.

## Diode Symbol

Figure 17(a) shows the basic diode structure and the standard schematic symbol for a general-purpose diode. The "arrowhead" in the symbol points in the direction opposite the electron flow. The two terminals of the diode are the anode (A) and cathode (K). The **anode** is the *p* region and the **cathode** is the *n* region.

When the anode is positive with respect to the cathode, the diode is forward-biased and current ($I_F$) is from cathode to anode, as shown in Figure 17(b). Remember that when the diode is forward-biased, the barrier potential, $V_B$, always appears between the anode and cathode, as indicated in the figure. When the anode is negative with respect to the cathode, the diode is reverse-biased, as shown in Figure 17(c) and there is no current. A resistor is not necessary in reverse bias but it is shown for consistency.



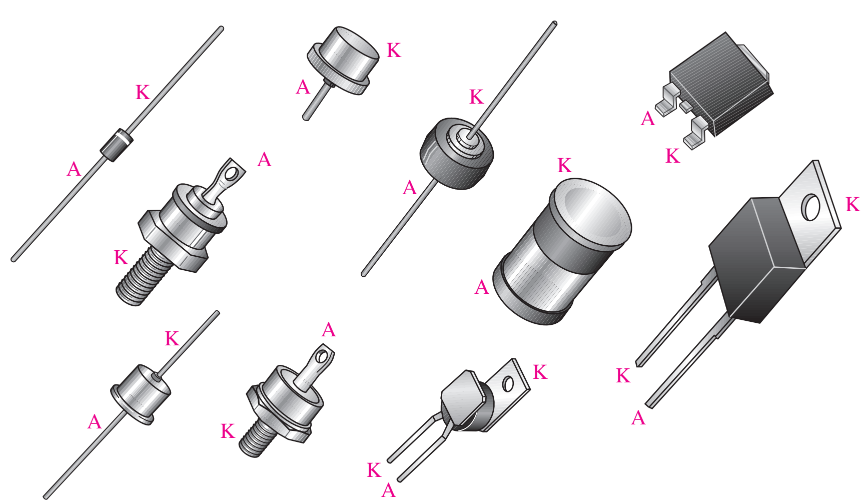

▲ **FIGURE 17**

**Diode structure, schematic symbol, and bias circuits. $V_{BIAS}$ is the bias voltage, and $V_B$ is the barrier potential.**

Some typical diodes and their terminal identifications are shown in Figure 18 to illustrate the variety of physical structures.

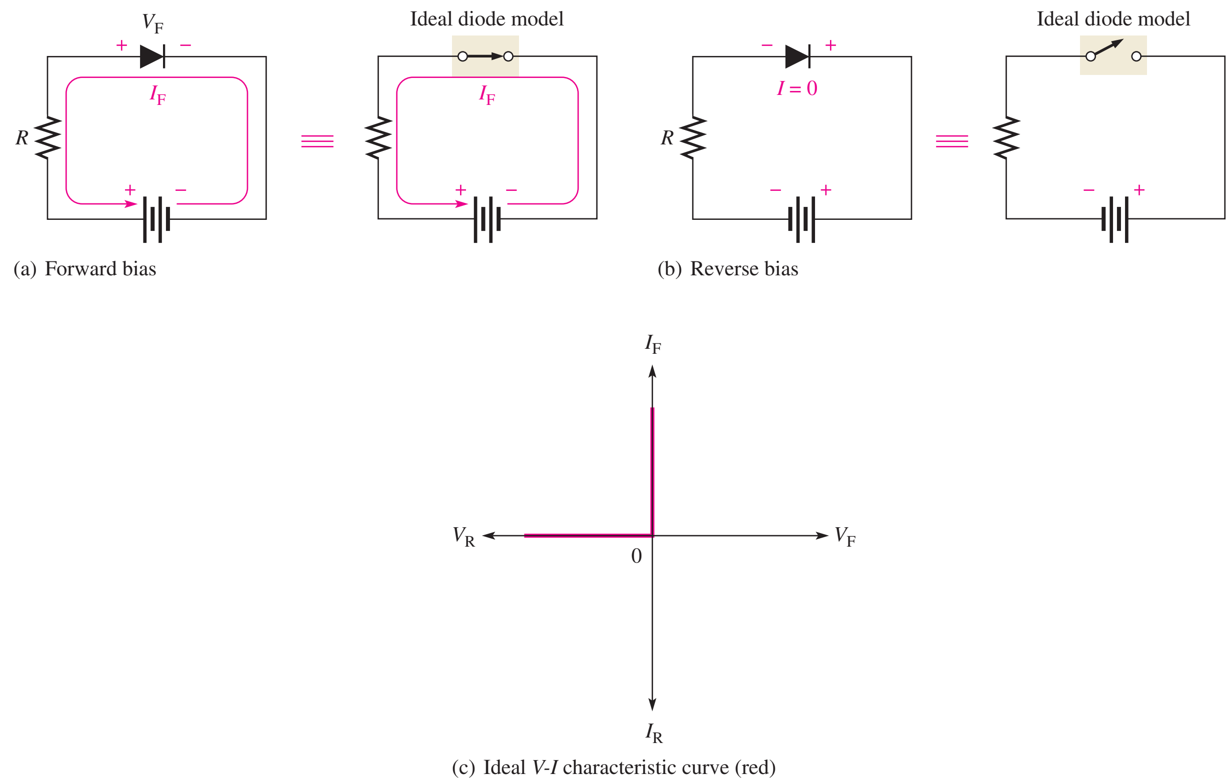
▶ **FIGURE 18**

**Typical diode packages and terminal identification. A is anode and K is cathode.**



## Diode Approximations

***The Ideal Diode Model*** The simplest way to visualize diode operation is to think of it as a switch. When forward-biased, the diode ideally acts as a closed (on) switch; and when reverse-biased, it acts as an open (off) switch, as shown in Figure 19. The *V-I* characteristic curve for this model is also shown in part (c). Note that the forward voltage ($V_F$) and the reverse current ($I_R$) are always zero in the ideal case.

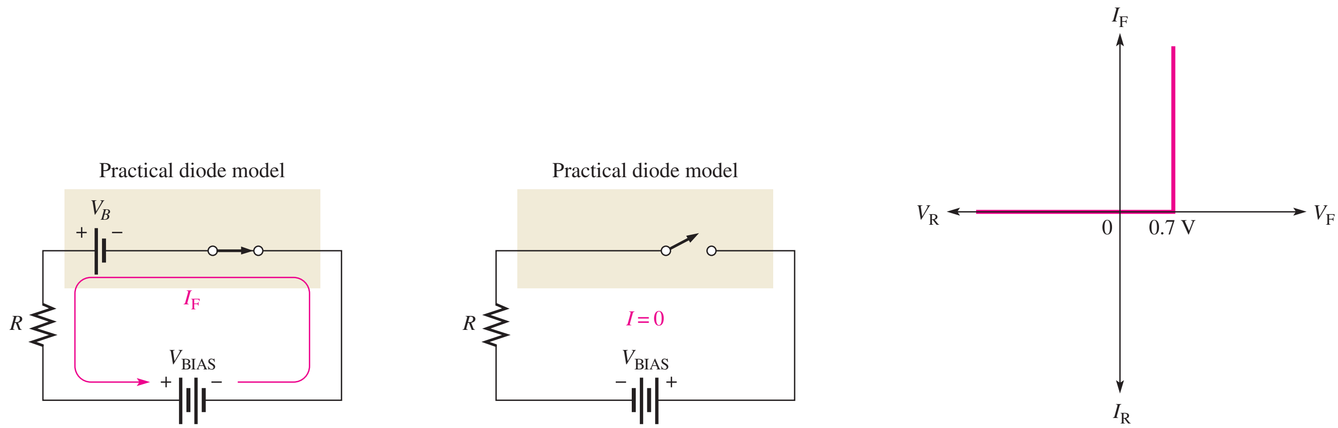



▲ **FIGURE 19**

**Ideal model of the diode as a switch.**

This ideal model, of course, neglects the effect of the barrier potential, the internal resistances, and other parameters. You may want to use the ideal model when you are troubleshooting or trying to figure out the operation of a circuit and are not concerned with more exact values of voltage or current.

***The Practical Diode Model*** The next higher level of accuracy includes the barrier potential in the diode model. In this approximation, the forward-biased diode is represented as a closed switch in series with a small "battery" equal to the barrier potential $V_B$ (0.7 V for Si), as shown in Figure 20(a). The positive end of the battery is toward the anode. *Keep in mind that the barrier potential only has the effect of a battery when forward bias is applied because the forward bias voltage, $V_{BIAS}$, must overcome the barrier potential before the diode begins to conduct current.* The reverse-biased diode is represented by an open switch, as in the ideal case, because the barrier potential does not affect reverse bias, as shown in Figure 20(b). The *V-I* characteristic curve for this model is shown in Figure 20(c).



(a) Forward bias (b) Reverse bias (c) *V-I* characteristic curve

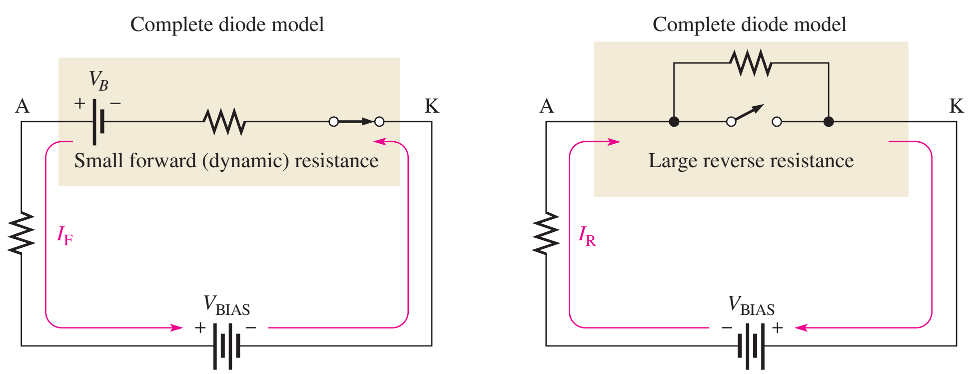
▲ **FIGURE 20**

The practical model of a diode.

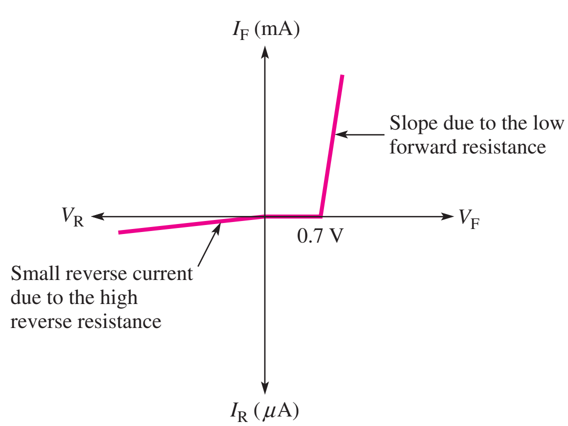
***The Complete Diode Model*** One more level of accuracy will be considered at this point. Figure 21(a) shows the forward-biased diode model with both the barrier potential and the low forward dynamic resistance. Figure 21(b) shows how the high internal reverse resistance affects the reverse-biased model. The characteristic curve is shown in Figure 21(c).

► **FIGURE 21**

The complete diode model includes barrier potential, forward resistance, and reverse resistance.



(a) Forward bias (b) Reverse bias



(c) *V-I* characteristic curve